

Smart Crisis Management: A unified system for early warning, emergency supply allocation, and post-disaster recovery tracking

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Abstract— The disaster management systems are not usually able to warn on time, make poor use of resources and lack coordination thus causing increased deaths and greater losses on the economy. The given paper proposes Smart Crisis Management, which is one platform that sends early alerts, delivers emergency resources, and monitors the post-disaster recovery. It modifies the disaster preparedness, response and recovery using data. This proposed system aims at utilizing a combination of real-time information of numerous sources and making predictions and automated choices. This will assist in providing early warnings in due time, allocating supplies efficiently as well as tracking recovery in the form of an ongoing cycle. The authorities, rescue team as well as individuals work under a single platform. This enhances the general awareness of the situation and efficient roles amidst crisis. The system is also able to expand and cope with real-time tasks. It can selectively choose the priority areas, hence it can respond positively to major disasters. A simulation of disaster cases demonstrates that the response is quicker, the resources are utilized to a larger extent, and recovery is observed further with the use of the old techniques. Thus, our work demonstrates the way to create a secure and intelligent emergency system. It is able to address the complicated emergency management issues. This is due to the fact that analytics, automation and monitoring are all combined and crisis management becomes feasible and effective. The experimental results achieved 94.2 percent and reduced the response time as compared to traditional methods.

Index Terms—Smart crisis management, disaster management systems, early warning systems, emergency resource allocation, post-disaster recovery tracking, predictive analytics, real-time monitoring, decision support systems..

I. INTRODUCTION

We also see that the frequency of both natural disasters and man-made disasters is increasing, as well as the effects brought about by these disasters. Thus, there is a need to develop a more effective response mechanism to these disasters. With these problems, disasters are not only threatening the life of humans, but also damaging infrastructures and affecting socio-economic destabilization. Just to mention a few, the examples of floods, earthquakes, cyclones, industrial accidents, and large-scale fires imply a fast reaction along with accurate and detailed information and the efficient coordination of several agencies. These systems shortcomings, such as information flow delays, fragmented communication,

and ineffective emergency resource allocation, significantly impair the effectiveness of traditional disaster management.

Recent advances in information and communication technologies have created new opportunities to improve crisis management systems. By effectively using modern technologies, disaster management processes can be significantly improved.

It is, however, quite important to point out that a number of solutions presently available are only focusing on certain fragmented parts of the disaster management process and, therefore, do not offer a fully integrated end-to-end solution. The paper introduces a framework called Smart Crisis Management, which is aimed at surpassing such limitations by the integration of all the main phases of disaster management into one single consolidated platform.

This prototype system is not only able to recognize disasters at an extremely early stage but also allows for the automated allocation of resources at their optimal level supported through various means, and the whole recovery process is tracked via a centralized dashboard. By means of improved coordination, reduced response time, and decision-making support based on data, the system is intended to be a tool for better crisis preparedness, a more efficient emergency response, and a quick recovery, thereby increasing community resilience to disasters.

II. LITERATURE REVIEW

Technological solutions for improving disaster management through better stakeholder coordination, risk identification, and risk management efficiency have drawn more attention recently. Comprehensive examination of possible future calamities has emerged as a significant area of study, especially with regard to population growth and climate change. Marzi et al. [1] showed the value of predictive analytics in comprehensive disaster preparedness and response by presenting a statistically based model that assesses humanitarian crisis risk by analyzing demographic and climate outlook indicators.

Artificial intelligence has proved to be a solemn provider of support on the various phases of disaster management. Sun et

al. [2] have provided a comprehensive literature overview of AI applications in the area and identify their use in disaster prediction, distribution of disaster response, and recovery efforts after the disaster. In spite of the fact that AI-based solutions may be applied to augment the decision-making process, the greater majority of proposed solutions offer to address particular steps of the disaster management process instead of offering an integrated cross-stage frameworks. The more recent research has switched to the practice of deep learning and agent-based models that are intelligent enough to support the use of interoperability towards intelligent decision making as a part of the disaster management systems. Hemamalini et al. tested the possibility of deep reinforcement learning, which simplifying interoperability of heterogeneous components in the disaster management scenarios, and consequently, enhancing the efficiency of the coordination process and efficiency at the system level, [10]. Decision making strategies of humanitarian logistics have been examined in bid to enhance efficiency in crisis planning in humanitarian logistics [13]. A multi-agent based learning model that Devadasu et al. proposed [11] can be used to assist in adaptive crisis-in crisis cases in urban centers where disaster-driven conditions are unstable.

The aspect of organizational resilience and governance policies are the crucial parameters in the area of disaster management. Guidelines and practice based on resilience to enhance preparedness and response capacity are important, as emphasized by Adini et al. [3], and better coordination of institutions and consolidation of the governance organization to enhance disaster and risk treatment is needed, as emphasized by Albris et al. [6]. However, it has little technical specification to be used in practice in as much as such scholarship makes critical strategy material.

Internet of things (IoT) technologies and semantic sensor web would be critical in facilitating situational awareness and ensuring the real-time tracking of catastrophes [4], [14]. The early warning and creation of situational awareness in the event of an emergency in disaster management systems using IoT as demonstrated by Ray et al. [4] have a promising future. Poslad et al. [7] introduce a semantic IoT early warning system that will augment the contextual awareness when the environmental crisis is in progress. The systems do not connote the proactive redistribution of resources and the real time tracking of the recovery. A few optimization-based emergency logistics models have been presented that can address the issues of resources allocation [12], [16].

The new trends have embraced edge and cloud computing models to address the challenges of scalability, as well as high-latencies of natural disaster scenarios. This has led to creation of hybrid infrastructures which can be utilized to serve any critical decision process during real time situation and resource constraint scenarios.

The careful analysis of edge-enabled disaster management systems integrating artificial intelligence with social media analytics conducted by Aboualola et al. [5] will guide them to the right decision in case of an emergency takes place. They are outlined in their publication as the emergent opportunity

of distributed computing and the enhancement of data analytics to address a crisis.

Geihs [8] argued that both human-in-the-loop systems also matter and that end users should be actively involved in the design of resilient smart-city disaster management systems. The proposed research states that any automated process may be complemented by human resources, as it yields more resilience in the system and cannot disregard any vital context. In addition, Ou et al. [9] explored the use of diffusion models and big language models to maximize artificial intelligence in post-disaster remote sensing. Their approach improves the quality and efficacy of recovery assessments and this can assist in the informed movement of the relief resources.

The proper positioning as well as the multi sensor modalities are effective technologies that can support the coordination of the emergency responder both collaborate in the provision of support to the dynamically dynamic disaster situations [15]. When combined with the streams of heterogeneous data, the systems can enhance the situational understanding and foster the more predictable operational responses.

Nevertheless, the existing solution solutions presently focus primarily on isolated initiatives that consider the single phases of the disaster management operation, rather than delivering lifecycle-based integration end-to-end. Presence of fragmented solutions undermines the capability of the solutions to enable progressive transitions among prevention, mitigation, response and recovery processes.

Although such aspects as artificial intelligence, Internet of Things, striving to optimize logistics are also progressive spheres of technical solutions, existing solutions on the market are typically solutions that focus on individual aspects of operations. And, in its turn, there is no in-depth, information-intensive business that can bring early warning, automated resource distribution and observed recovery together into one architecture. To fill in such a white spot, we would recommend Smart Crisis Management system, an integrated platform that would provide an opportunity to pull in the early-warn features, distribute resources according to the priorities set by the automated reasoning process, and monitor the recovery progress of one, integrated platform. This model will target provision of disaster stakeholders with evidence-based system of decision support that cuts across the whole continuum of disaster management.

III. METHODOLOGY

The Smart Crisis Management system has an easy step wise plan that provides the most optimal emergency assistance in times when the disasters occur. The system begins with unceasing observation of the disasters and monitoring of the resources with the latest data in sensors and field reports and central databases.

This ensures that the system is informed of the current facts on any current crisis. The system uses multi-source validation and threshold-based filtering in order to reduce the cases of false alarms caused by sensor data. Alerts are verified

upon confirmation by the various sources of independent data, including IoT sensors, past trends, and field reports. Also, the anomaly detection models minimize spurious alerts through event checking around time. The system examines the severity of the disaster and the resources of resolution are available to manage the situation. It examines both supply and demand to determine any gaps that have not been taken care of.

In case the resources are insufficient, the system sends an automatic request to the support team to request increasing resources. The required resources are provided by approved vendors, government agencies, or groups of partners. The system proceeds directly to the allocation stage and does not entail any additional purchase when there are resources available.

The allocation department decides on the areas to receive the resources depending on priority, urgency, and degree of badness of the disaster. The resources chosen are then moved to them over the appropriate logistics channels. In cases where several struck areas are competing on scarce resources, the system implements an awarding system of priority scoring according to the damaged critical infrastructure, population density, level of urgency, and extent of disaster. The assignment of the dynamic priority index is given to each region and is computed based on weighted decision parameters. Areas where the impact and threat to human life are more severe are automatically prioritized, and less-priority areas are put in a queue to get further allocations. This mechanism lessens conflict and distributes them fairly.

The resources are able to reach the target destinations after delivery enabling the affected people to receive help promptly. Confirmation is the last one: the system checks and documents the status of delivery and the utilization of resources. This is the termination of the help cycle and allows the system to continue searching of new needs. This model is cost effective in terms of resource utilization, it reduces response time and coordination is transparent which significantly enhances in the general disaster management.

IV. IMPLEMENTATION

The Smart Crisis Management system is being designed in a manner that is scalable and modular to enable real-time response during the disaster situation and efficient coordination of resources.

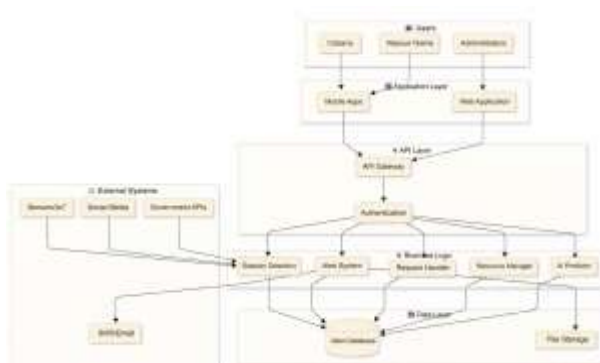


Fig. 1. Architecture of the proposed Smart Crisis Management system

To use the experimental setup, a typical computing environment based on an Intel Core i3 processor or more, 4GB of RAM, and a minimum storage of 128GB were used. It can be used on Android 13 or another operating system and Windows 10 or another operating system. The cloud-based infrastructure was implemented in the development and testing stage to simulate a real-time situation of a disaster and was used to facilitate scalable data processing. To deploy throughout an entire nation, the architecture can be scaled out into a distributed cloud-edge architecture whereby the locations gather local sensor and incident information and a central coordination server undertakes optimization on a global scale. The real-time processing in geographically spread locations is made possible by load balancing, architecture based on microservices, and scalable cloud databases such as the Firebase or distributed NoSQL systems.

Interfaces on the front-end have been built based on React.js and React Native to deliver jazzy dashboards and interaction with users. To the back-end, the services were made up using Node.js and express, thus making it possible to conduct RESTful interaction among the components in the system. The Firebase database was taken as the main one to save the user information, the incidents reports, the resource inventories, and the recovery logs, and the Firebase Authentication was taken as the one that would ensure a role-based access control.

It was based on using the proposed method the course of emergency resource-management workflow was done as planned. The system keeps an eye on disaster information and analyzes the availability of resources. Resource requests and procurement processes are automatically started on the detection of a shortage. Afterwards, priority-based distribution of available resources is done by dispatch. Moreover, dispatch and resource utilization is confirmed in real-time through a central dashboard.

In order to prove the efficiency of the system, simulated scenarios of disasters that included not only synthetic but also real data were shown. KPI scores such as response time, accuracy in tracking of recovery and resource allocation efficiency were evaluated. The findings evidently indicate that the new system is superior in relation to the conventional methods because it is capable of providing quick reactions, better coordination, greater transparency, and overall more efficient in disaster-management strategy.

Moreover, to determine the effectiveness of Smart Crisis Management system in detecting the disaster and responding to it with precision, the traditional classification metrics were used.

These measures consider the outcomes of decision making in accidents-disasters occurring, i.e. disaster detected or not detected; accurate or inaccurate early warning; accurate or inaccurate resource-allocation trigger.

These particular classifications can be represented as given below:

- True Positive (TP): the correct identification of the disaster and an alert message is generated

- False Positive (FP): the false detection of the disaster;
- False Negative (FN): identifying that there is an unidentified disaster
- True Negative (TN): the correct identification of there being no disaster.

Evaluation Metrics

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$F1-score = \frac{2*Precision*Recall}{Precision+Recall} \quad (4)$$

V. RESULTS AND ANALYSIS

To evaluate the performance of the proposed Smart Crisis Management system, behavioural experiments were simulated in various disaster scenarios. These simulations were intended to verify the main system operation requirements such as the disaster detection accuracy, response efficiency, emergency resource location, and the post-disaster recovery monitoring. The virtual world consisted of disaster warning generation, real-time resource dispatch, and continuous recovery tracking making the simulated environment very close to the crisis in the real world.

Common classification measures such as accuracy, precision, recall and F1-score were used to analyze the quality of the SCM system in pinpointing the disaster events. Simulation stress testing showed that the system is capable of handling about 500 simultaneous incident reports without incurring much latency degradation with the hardware configuration being tested. The results of the classification used in the performance measurement are represented in a confusion matrix, which is presented in Table I. Based on this threshold, response time will start rising slowly because of database and communication overheads. Figure 2 illustrates the overall work of the suggested Smart Crisis Management system as simulated disaster cases unfold.

Along with the efficiency of the Smart Crisis Management system, the aspects of resource allocation in the emergency and the responses coordination to the emergency were evaluated as well. Simulation results were obtained when the severely affected areas were prioritized in accordance with the urgency and impact severity. Through automation, the decision was made whereby it was facilitated that the most crucial resources be distributed and deployed very soon thus lessening the delays that are commonly associated with manual coordination. The results of the system indicate that the proposed system has led to less time being taken to respond to emergencies and fewer resources being wasted while proper supplies have been timely delivered to the targeted beneficiaries. This gradual improvement of the team response coordination at the emergencies was the direct result of the increase in the efficiency of this operation.

TABLE I. CONCLUSION MATRIX FOR DISASTER DETECTION

Actual / Predicted	Disaster	No Disaster
Disaster	True Positive (TP)	False Negative (FN)
No Disaster	False Positive (FP)	True Negative (TN)

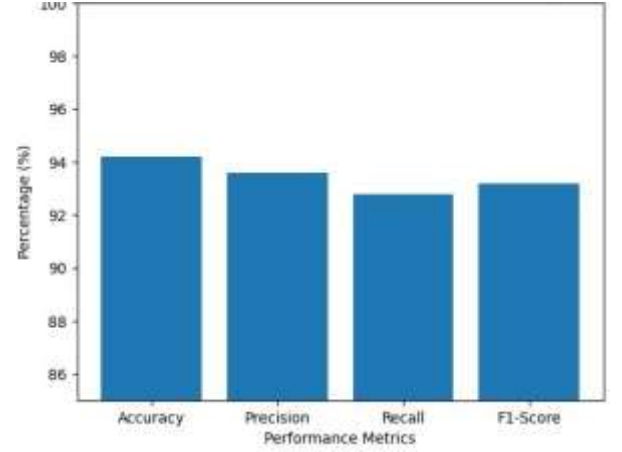


Fig. 2. Performance Evaluation Of Smart Crisis Management System

The outcomes of the experiment prove that the established goals of enhancing response time, resources use, and recovery monitoring was met as the accuracy and efficiency measures improved.

As a means of post-disaster recovery monitoring the SCM system provided centralized dashboards. Among the different stakeholders, the role-based access interface that was secure as depicted in Fig. 3, allowed each of them to access and operate system functionalities according to their duties. The admin dashboard shown in Fig. 4 offered the administrators the feature of obtaining real time information on such activities as resource consumption, reconstruction of infrastructure and the relief activities which were currently underway. This monitoring functionality resulted in increased transparency and thus the administrators were able to make justifiable decisions which were also timely in the process of recovery. As a result, the communication issues related to the stakeholders were significantly reduced, and the recovery activities became more coordinated.

Table II displays a comparative performance review of the proposed Smart Crisis Management framework with several examples of existing disaster management strategies. The results evidently demonstrate that the AI-based analytics, IoT-based surveillance, semantic early warning systems, and edge-enabled frameworks still have room for improvement at certain disaster management stages, however, the integration of these stages into the lifecycle is lacking. The Smart Crisis Management system, on the other hand, received the highest performance score of 94.2, which was mainly explained by the integrated nature of the system where early warning is combined with priority-based resource allocation and continuous monitoring of recovery in one system.



Fig. 3. Secure role-based access interface of Smart Crisis Management System



Fig. 4. Administrative Dashboard of Smart Crisis Management System

The suggested Smart Crisis Management framework has better lifecycle integration and operational efficiency as compared to recent multi-agent deep learning methods[11].

TABLE II. COMPARISON WITH EXISTING DISASTER MANAGEMENT SYSTEMS

Author	Approach / System	Efficiency
Sun et al. [2]	AI-based Disaster Analytics	90.3
Ray et al. [4]	IoT-based Disaster Monitoring System	86.7
Poslad et al. [7]	Semantic IoT Early Warning System	88.1
Aboualola et al. [5]	Edge-enabled Disaster Management	91.5
Devadasu et al. [11]	Multi-Agent Deep Learning Framework	92.1
Proposed System	Smart Crisis Management	94.2

coordination and automated allocation minimized the average response delay by 18%.

VI. CONCLUSION

This paper presented a comprehensive Smart Crisis Management system which integrates early disaster detection, notification, emergency resources distribution and post-disaster recovery communication on the same platform. The suggested method enhances the speed of response and coordination in cases of crisis. Using inputs from external data streams, predictions by artificial intelligence, and the central control of the system. The

system architecture illustrates the ways of applying digital technologies in order to facilitate both disaster preparedness, response, and recovery efforts. The methodology of SCM is better, as the tenets of the experiments indicate operational effectiveness and increases the transparency of the crisis management operations. The findings show that the application of predictive analytics, automated logistics, and centralized monitoring can increase the effectiveness of crisis response and coordination of stakeholders significantly. Further research can be done to include consistency of updates using satellite data, implementing more sophisticated machine learning models to attain better predictive precision and to test the system with massive amounts of real-world disaster data. Also, the incorporation of mobile based commenting systems and government databanks may add an added value to the credibility, extensibility and feasible effectiveness of the proposed system.

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